

Adrianna Jerrell Milton, PhD

Email: Adrianna.j.milton@dartmouth.edu

Phone: +1 (786) 267-7607

Education

Doctor of Philosophy, Neuroscience

August 2023

Case Western Reserve University, Cleveland, Ohio, U.S.A.

Bachelor of Science, Psychology

May 2014

Florida State University, Tallahassee, Florida, U.S.A.

Employment

Mindfulness Instructor

2024-Present

Student Wellness Center, Dartmouth College, Hanover, New Hampshire, U.S.A.

High School Research Mentor

2024-Present

Polygence

Postdoctoral Researcher

September 2023-June 2026

Department of Molecular and Systems Biology, Geisel School of Medicine, Dartmouth College, Hanover, New Hampshire, U.S.A.

Post-baccalaureate Research Education Program

June 2014-June 2015

Center for Addiction Sciences and Therapeutics, University of Texas Medical Branch, Galveston, Texas, U.S.A.

Teaching Experience

Course Assistant

Summer 2026

Dartmouth Precollege, Dartmouth College, Hanover, New Hampshire, U.S.A.

Summer Scholars Program: Psychology and Cognitive Neuroscience course

Invited Guest Lecturer

Spring 2026

Dartmouth Center for the Advancement of Learning, Dartmouth College, Hanover, New Hampshire, U.S.A.

Learning Community for Future Faculty: “Mindfulness in Academia: Promoting Wellness for Academic Success.”

Invited Guest Lecturer

Spring 2026

Student Wellness Center, Dartmouth College, Hanover, New Hampshire, U.S.A.
Food and Body Speaker series: “Following Your Gut: Hunger, Interoception, and Mindful Eating.”

Invited Guest Lecturer

Spring 2026

Dartmouth Institute for Health Policy and Clinical Practice, Dartmouth College, Hanover, New Hampshire, U.S.A.

Public Health Webinar Series: “Molecular investigations of neurobiology in zebrafish.”

Guest Lecturer

Spring 2025

Psychological and Brain Sciences, Dartmouth College, Hanover, New Hampshire, U.S.A.

Neurobiology of Fear and Anxiety Undergraduate Course: “Left habenular activity attenuates fear responses in larval zebrafish.”

Provost Scholar Tutor

2019-2020

Provost Scholars Program, Case Western Reserve University, Cleveland, Ohio, U.S.A.

Regularly met one-on-one with high school students from area of low socioeconomic status weekly for tutoring sessions on various subjects.

Teaching Assistant

January 2014-May 2014

Physiological Psychology Lab, Florida State University, Tallahassee, Florida, U.S.A.

Instructed undergraduate students on gross neuroanatomy using preserved sheep brain samples. Administered and graded corresponding assignments and examinations to assess appropriate application of coursework.

Tutor and Mentor

August 2013-May 2014

Center for Academic Retention and Enhancement, Florida State University, Tallahassee, Florida, U.S.A.

Served as an educational counselor, mentor, and tutor to underrepresented minorities and economically at-risk undergraduate students. Tutored in subjects including math, biology, English, psychology, pharmacology, and neuroscience.

Research Experience

Postdoctoral Researcher

September 2023-June 2026

Dartmouth Geisel School of Medicine, Hanover, New Hampshire, U.S.A.

Department of Molecular and Systems Biology

Utilizes transgenic and optogenetic methods in the zebrafish model to neuroanatomically map and functionally manipulate neural pathways during behavior.

Mentor: Dr. Marnie Halpern

Graduate Student Researcher

May 2017-August 2023

Case Western Reserve University, Cleveland, Ohio, U.S.A.

Department of Neuroscience, School of Medicine

Surgically injured rats in a model of chronic cervical spinal cord injury to test systemic treatment(s) aimed at restoring forelimb mobility.

Mentor: Dr. Jerry Silver

Graduate Student Researcher**January 2016–April 2017**

Case Western Reserve University, Cleveland, Ohio, U.S.A.

Department of Neuroscience, School of Medicine

Devised, built, and implemented an automated behavioral paradigm to measure odorant preference in wild-type mouse lines to assess ecological relevance. Employed in vivo electrophysiological techniques to record and analyze neuronal potential within deep, internal brain structures.

Mentor: Dr. Daniel Wesson**Research Assistant as P.R.E.P Scholar****June 2014–June 2015**

University of Texas Medical Branch, Galveston, Texas, U.S.A.

Department of Pharmacology and Toxicology

Evaluated learning and memory using behavioral memory assay Morris Water Maze in transgenic mouse line. Employed immunohistochemical techniques to visualize and measure localized neural protein expression using fluorescent and confocal microscopy.

Mentor: Dr. Jonathan Hommel**Research Assistant****May 2013–May 2014**

Florida State University, Tallahassee, Florida, U.S.A.

Department of Neuroscience

Reliably operated automated behavioral gustometer assays to measure and analyze taste detection and discrimination in gastric bypass rats and a transgenic knockout mouse line to understand the sensory transduction of taste perception.

Mentor: Dr. Alan Spector**Research Assistant****January 2013–January 2014**

Florida State University, Tallahassee, Florida, U.S.A.

Department of Psychology

Worked directly with testing volunteers to clearly communicate and administer computerized testing methods designed to analyze memory consolidation and retrieval in enriched context to understand the role of emotion during these processes.

Mentor: Dr. Colleen Kelley**Publications**

Milton, A. J., Kwok, J. C. F., McClellan, J., Randall, S. G., Lathia, J. D., Warren, P. M., Silver, D. J., & Silver, J. (2023). Recovery of Forearm and Fine Digit Function After Chronic Spinal Cord Injury by Simultaneous Blockade of Inhibitory Matrix Chondroitin Sulfate Proteoglycan Production and the Receptor PTP σ . *Journal of neurotrauma*, 40(23-24), 2500–2521. <https://doi.org/10.1089/neu.2023.0117>

Midroit, M., Chalençon, L., Renier, N., **Milton, A.**, Thevenet, M., Sacquet, J., Breton, M., Forest, J., Noury, N., Richard, M., Raineteau, O., Ferdenzi, C., Fournel, A., Wesson, D. W., Bensafi, M., Didier, A., & Mandairon, N. (2021). Neural processing of the reward value of pleasant odorants. *Current biology : CB*, 31(8), 1592–1605.e9. <https://doi.org/10.1016/j.cub.2021.01.066>

Gadziola, M. A., Stetzik, L. A., Wright, K. N., **Milton, A. J.**, Arakawa, K., Del Mar Cortijo, M., & Wesson, D. W. (2020). A Neural System that Represents the Association of Odors with Rewarded Outcomes and Promotes Behavioral Engagement. *Cell reports*, 32(3), 107919. <https://doi.org/10.1016/j.celrep.2020.107919>

Jagetia, S., **Milton, A. J.**, Stetzik, L. A., Liu, S., Pai, K., Arakawa, K., Mandairon, N., & Wesson, D. W. (2018). Inter- and intra-mouse variability in odor preferences revealed in an olfactory multiple-choice test. *Behavioral neuroscience*, 132(2), 88–98. <https://doi.org/10.1037/bne0000233>

Kasper, J. M., McCue, D. L., **Milton, A. J.**, Szwed, A., Sampson, C. M., Huang, M., Carlton, S., Meltzer, H. Y., Cunningham, K. A., & Hommel, J. D. (2016). Gamma-Aminobutyric Acidergic Projections From the Dorsal Raphe to the Nucleus Accumbens Are Regulated by Neuromedin U. *Biological psychiatry*, 80(11), 878–887. <https://doi.org/10.1016/j.biopsych.2016.02.031>

Conferences and Presentations

Molecular and Systems Biology Department Retreat **September 2025**
Poster: “Transsynaptic tracing in zebrafish: challenges and new applications.”

Summer Program in Neuroscience, Excellence and Success (SPINES) **Summer 2025**
Marine Biological Laboratory at Woods Hole, MA
Datablitz presentation: “Transsynaptic mapping of the zebrafish gut-brain axis.”

DartFISH at Dartmouth **January 2025**
Seminar: “Transsynaptic mapping of the zebrafish gut-brain axis.”

NEURAL Conference **June 2023**
University of Alabama at Birmingham, AL
Oral presentation: “Recovery of forearm and fine digit function after chronic spinal cord injury by simultaneous blockade of inhibitory matrix CSPG production and the receptor PTP σ .”

Seminar Invited Speaker **November 2022**
Provost Scholars, Case Western Reserve University, Cleveland, OH
Oral presentation: “Collectively Individual”

45th Annual Biomedical Graduate Student Symposium **November 2022**
Case Western Reserve University, Cleveland, OH
Oral presentation: “Recovery of forearm and fine digit function after chronic spinal cord injury by simultaneous blockade of inhibitory matrix CSPG production and the receptor PTP σ .”

Society for Neuroscience **November 2022**
San Diego, CA
Poster: “Recovery of forearm and fine digit function after chronic spinal cord injury by simultaneous blockade of inhibitory matrix CSPG production and the receptor PTP σ .”

Seminar Invited Speaker **October 2022**
Dartmouth Geisel School of Medicine, Hanover, NH

Oral presentation: “Recovery of forearm and fine digit function after chronic spinal cord injury by simultaneous blockade of inhibitory matrix CSPG production and the receptor PTP σ .”

Neuroscience Departmental Retreat

July 2022

Case Western Reserve University, Cleveland, OH

Oral presentation: “Recovery of forearm and fine digit function after chronic spinal cord injury by simultaneous blockade of inhibitory matrix CSPG production and the receptor PTP σ .”

NEURAL Conference

June 2019

University of Alabama at Birmingham, AL

Poster: “Manipulating perineuronal net inhibition to induce recovery of forelimb function after acute and chronic cervical spinal cord injury.”

NEURAL Conference

June 2018

University of Alabama at Birmingham, AL

Poster: “Overcoming and Degrading Inhibitory Proteoglycans Globally to Promote Axon Regeneration and Functional Recovery after a Chronic Thoracic Spinal Cord Injury.”

Society for Neuroscience

November 2018

San Diego, California, U.S.A.

Poster: “Overcoming and Degrading Inhibitory Proteoglycans Globally to Promote Axon Regeneration and Functional Recovery after a Chronic Thoracic Spinal Cord Injury.”

Society for Neuroscience

October 2015

Chicago, Illinois, U.S.A.

Poster: “Adipocyte-specific overexpression of ecto-nucleotide pyrophosphate phosphodiesterase-1 leads to memory impairment in mice on a high-fat diet.”

Honors/ Awards

Samuel Ford McGill Award (Finalist)

January 2026

The Office of Diversity, Inclusion & Community Engagement (DICE), Dartmouth College, Hanover, New Hampshire, U.S.A.

Best Oral Presentation (3rd Place)

June 2023

NEURAL Conference, University of Alabama at Birmingham, Birmingham, Alabama, U.S.A.

Shark Tank Presentation (2nd Place)

June 2019

NEURAL Conference, University of Alabama at Birmingham, Birmingham, Alabama, U.S.A.

Who’s Who in Black Cleveland 14th Edition

June 2019

Poster Presentation (2nd Place)

June 2018

NEURAL Conference, University of Alabama at Birmingham, Birmingham, Alabama, U.S.A.

Psi Chi Honor Society Chapter

Fall 2013

Volunteer of the Year, Florida State University, Tallahassee, Florida, U.S.A.

Dean's List

Fall 2011

Florida State University, Tallahassee, Florida, U.S.A.

Travel Awards

NEURAL Conference

June 2023

University of Alabama at Birmingham, Birmingham, Alabama, U.S.A.

NEURAL Conference

June 2019

University of Alabama at Birmingham, Birmingham, Alabama, U.S.A.

NEURAL Conference

June 2018

University of Alabama at Birmingham, Birmingham, Alabama, U.S.A.

Association for Chemoreception Sciences Conference

March 2017

Bonita Springs, Florida, U.S.A.

Fellowships, Grants and Certifications

Constructive Dialogue Facilitator

Spring 2026

Constructive Dialogue Institute

Unity Banquet Scholarship

Spring 2019

Office of Multicultural Affairs, Case Western Reserve University, Cleveland, Ohio, U.S.A.

Weatherhead Executive Education Program Grant

Spring 2019

School of Graduate Studies Professional Development Center, Case Western Reserve University, Cleveland, Ohio, U.S.A.

Mentorship

Graduate Mentor

Spring 2025-Spring 2026

Presidential Scholars Program, Dartmouth College, Hanover, New Hampshire, U.S.A.

Provided laboratory learning environment and practical, hands-on scientific mentorship for an undergraduate student from Dartmouth, New Hampshire, U.S.A.

Undergraduate mentee: Michael Humphreys

Graduate Mentor

Summer 2024

Summer Undergraduate Research Fellowship (iSURF), Dartmouth College, Hanover, New Hampshire, U.S.A.

Provided laboratory learning environment and practical, hands-on scientific mentorship for an undergraduate student from Plymouth State University, New Hampshire, U.S.A.

Undergraduate mentee: Becket White

Graduate Mentor**Fall 2020-Summer 2023**

Student Internship, Case Western Reserve University, Cleveland, Ohio, U.S.A.

Provided laboratory learning environment and practical, hands-on scientific mentorship for an undergraduate student from Case Western Reserve University, Cleveland, Ohio, U.S.A.

Undergraduate mentee: Carly Tucci

Graduate Mentor**Fall 2021-Summer 2022**

Student Internship, Case Western Reserve University, Cleveland, Ohio, U.S.A.

Provided laboratory learning environment and practical, hands-on scientific mentorship for an undergraduate student from Case Western Reserve University, Cleveland, Ohio, U.S.A.

Undergraduate mentee: Gerardo Pena

Graduate Mentor**Spring 2022-Fall 2022**

Student Internship, Case Western Reserve University, Cleveland, Ohio, U.S.A.

Provided laboratory learning environment and practical, hands-on scientific mentorship for an undergraduate student from Case Western Reserve University, Cleveland, Ohio, U.S.A.

Undergraduate mentee: Maheen Ahmed

Provost Scholar Mentor**2021-2023**

Provost Scholars Program, Case Western Reserve University, Cleveland, Ohio, U.S.A.

Planned and met weekly to engage in personal development activities with a female, minority high school student from an underserved Cleveland school district.

11-12th grade mentee: Sha'Niya Jackson

Graduate Mentor**Summer 2021-Summer 2022**

Student Internship, Case Western Reserve University, Cleveland, Ohio, U.S.A.

Provided laboratory learning environment and practical, hands-on scientific mentorship for an undergraduate student from Case Western Reserve University, Cleveland, Ohio, U.S.A.

Undergraduate mentee: Shweta Mohapatra

Graduate Mentor**Summer 2021**

Student Internship, Case Western Reserve University, Cleveland, Ohio, U.S.A.

Provided laboratory learning environment and practical, hands-on scientific mentorship for an undergraduate student from Case Western Reserve University, Cleveland, Ohio, U.S.A.

Undergraduate mentee: Priyanka Agochiya

Graduate Mentor**Summer 2019**

Student Internship, Case Western Reserve University, Cleveland, Ohio, U.S.A.

Provided laboratory learning environment and practical, hands-on scientific mentorship for a female, minority high school student from Hawken School, Cleveland, Ohio, U.S.A.

12th grade mentee: Jordyn Singleton

Graduate Mentor**Summer 2019**

Student Internship, Case Western Reserve University, Cleveland, Ohio, U.S.A.

Provided laboratory learning environment and practical, hands-on scientific mentorship for a female, minority high school student from Hawken School, Cleveland, Ohio, U.S.A.

12th grade mentee: Cordelia Zheng

Graduate Mentor**Fall 2018-Spring 2022**

Student Internship, Case Western Reserve University, Cleveland, Ohio, U.S.A.

Provided laboratory learning environment and practical, hands-on scientific mentorship for an undergraduate student from Case Western Reserve University, Cleveland, Ohio, U.S.A.

Undergraduate mentee: John Barnett**Graduate Mentor****Summer 2018**

Northern Ohio AGEF Alliance, Case Western Reserve University, Cleveland, Ohio, U.S.A.

Provided laboratory learning environment and practical, hands-on scientific mentorship for a female, minority undergraduate student from Fisk University, Nashville, Tennessee, U.S.A. The Northern Ohio AGEF Alliance (NOA-AGEF) is an alliance of seven northern Ohio universities developed to ensure that underrepresented minority students receive additional support and professional development.

Undergraduate mentee: Anastacia Vance**Graduate Mentor****Summer 2018**

Student Internship, Case Western Reserve University, Cleveland, Ohio, U.S.A.

Provided laboratory learning environment and practical, hands-on scientific mentorship for a female, minority high school student from Hawken School, Cleveland, Ohio, U.S.A.

11th grade mentee: Jordyn Singleton**Graduate Mentor****Summer 2018**

Scientific Enrichment Opportunity Program, Case Western Reserve University, Cleveland, Ohio, U.S.A.

Provided laboratory learning environment and practical, hands-on scientific mentorship for a female high school student from Berea High School, Cleveland, Ohio, U.S.A. The Scientific Enrichment Opportunity Program offers high school students' exposure to biomedical laboratory settings to gain a first-person perspective of the scientific process and atmosphere of critical thinking, as well as inspire the confidence to pursue a college degree.

11th grade mentee: Kortney Mave**Graduate Mentor****Fall 2017-Summer 2018**

Student Internship, Case Western Reserve University, Cleveland, Ohio, U.S.A.

Provided laboratory learning environment and practical, hands-on scientific mentorship for an undergraduate student from Case Western Reserve University, Cleveland, Ohio, U.S.A.

Undergraduate mentee: Evelyn de Groot**Professional and Scientific Memberships**

Council on the Libraries at Dartmouth

*Postdoctoral representative***2025-2026**

Dartmouth Postdoctoral Association (DPA)

*Co-President***2025-2026***Board Member***2024-2025**

Society for Neuroscience - Neuroscience Scholars Program (NSP)

*NSP Associate***2025-Present**

STEM Programs for Aspiring Researchers and Kids (S.P.A.R.K.) @ Dartmouth <i>Vice President for Education</i>	2023-2024
The Dartmouth Black Caucus	2023-Present
Boston Area Zebrafish Researchers (BAZaR)	2023-Present
National Center for Faculty Development & Diversity (NCFDD)	2023-Present
DartFISH at Dartmouth	2023-2026
Black In Neuro	2022-Present
Society for Neuroscience	2015-2023

Professional and Community Service

Event Volunteer

Dartmouth College, Hanover, New Hampshire, U.S.A.	
11 th Annual Science Day at Dartmouth, Guarini GRAD	Spring 2026
12 th Annual Upper Valley Brain Bee, Integrative Neuroscience at Dartmouth	Spring 2026
Levey Engineering Open House at Dartmouth, Dartmouth Engineering	Spring 2026
Plum Village Monastics Mindfulness Retreat, Mindful Dartmouth	Spring 2026
Launch the NEXT Frontier, Dartmouth NEXT	Winter 2026

Planning Committee

Dartmouth Postdoc Association, Dartmouth College, Hanover, New Hampshire, U.S.A.	
Postdoc Research Day	Winter 2026

Planning Committee Programming Lead

Hopkins Center for the Arts at Dartmouth, Hanover, New Hampshire, U.S.A.	
Winter Mindfulness Event	Winter 2025

Mindfulness Eating Event Co-Facilitator

Geisel School of Medicine, Guarini GRAD, House Communities, and the Student Wellness Center Dartmouth College, Hanover, New Hampshire, U.S.A.	
Summer Community Mindful Lunch	July 2025
National Postdoc Appreciation Week Mindful Lunch	September 2025
Fall Community Mindful Lunch	October 2025
Thanks Giving Mindful Dinner	November 2025
Winterim Community Mindful Dinner	December 2025
Spring Break Community Mindful Lunch	March 2026
Summer Community Mindful Picnic	July 2026

Trainee Panelist

Hopkins Center for the Arts at Dartmouth College, Hanover, New Hampshire, U.S.A. The Buddha the Scientist Symposium <i>Recording available here</i>	March 2025
S.T.E.M. Outreach with S.P.A.R.K. @ Dartmouth Ottauquechee School, White River Junction, Vermont, U.S.A. Community Science Night	2023-2026
S.T.E.M. Outreach with S.P.A.R.K. @ Dartmouth Middlebury Middle School, Middlebury, Vermont, U.S.A. Girls S.T.E.M. Club	2023-2024
Planning Committee Case Western Reserve University, Cleveland, Ohio, U.S.A. Neurosciences Department 2022 Retreat	Spring 2022
Planning Committee University of Alabama at Birmingham, Birmingham, Alabama, U.S.A. NEURAL Conference	Summer 2020
Science Fair Keynote Speaker East Cleveland School District, Cleveland, Ohio, U.S.A.	January 2020

References

1. Marnie Halpern, PhD
Chair, Department of Molecular and Systems Biology
Geisel School of Medicine, Dartmouth College
Marnie.E.Halpern@dartmouth.edu
2. Heather Broihier, PhD
Professor, Department of Neurosciences
School of Medicine, Case Western Reserve University
Heather.Broihier@case.edu
3. Alicia Battle, PhD, MCHES, MS
Assistant Dean for Public Health Programs and Assistant Professor
The Dartmouth Institute for Health Policy and Clinical Practice, Geisel School of Medicine,
Dartmouth College
Alicia.L.Battle@dartmouth.edu